

RESEARCH ARTICLE

A Theoretical Architecture for Cyclic Prosodification

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1. Introduction

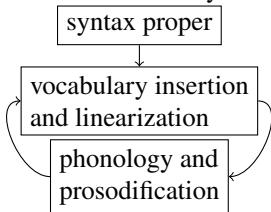
In recent years, the literature on the syntax-prosody interface is characterized by a heated debate about whether an independent notion of a prosodic constituent is necessary or not. On the one hand, there are the so-called Direct Mapping approaches which assume that notions like a prosodic word or a prosodic phrase always correspond in some sense to morphosyntactically established domains and are therefore redundant (see amongst many others D'Alessandro & Scheer 2015; Dobashi 2003; Fenger 2020; Fenger & Weisser accepted; Kaisse 1985; Newell 2008; Newell & Piggott 2014; Pak 2008; Wagner 2005). On the other hand, there is the so-called Indirect Mapping approaches, which emphasize

that we find plenty of cases where prosodic constituency cannot readily be reduced to morphosyntactic constituency (cases of so-called non-isomorphism) and that we do need an independent ontological notion of a prosodic hierarchy and the elements it contains (see amongst many others [Cheng & Downing 2016](#); [Elfner 2012, 2015](#); [Nespor & Vogel 1986](#); [Selkirk 1982, 2000, 2009](#); [Truckenbrodt 1999](#)).

Interestingly, these approaches do not only differ in whether they assume prosodic constituents being required as independent ontological notions but they also, at least in the vast majority of the cases, differ as to whether they assume the syntax-prosody mapping to apply cyclically or not. Influenced by cyclicity effects in structure-building modules syntax and morphology, Direct Mapping approaches typically apply in a cyclic fashion with the equivalents of prosodic constituents being built in a cyclic, bottom-up way. Indirect Mapping approaches typically map entire syntactic trees (or at least big chunks of syntactic structure) to prosodic constituents in a representational way.¹ But even though the issue of cyclicity is an important one as it makes radically different predictions, it is rarely discussed. Thus, it is precisely this dimension of cyclicity that we want to focus on in this paper.

We put forward a strictly cyclic model of the syntax-prosody interface in which syntactic structures are prosodified in a stepwise fashion as soon as the respective terminal nodes receive phonological content, i.e. during Vocabulary Insertion. As Vocabulary Insertion applies bottom-up (see e.g. [Bobaljik \(2000, 2012\)](#); [Embick \(2010\)](#); [Kalin \(2022\)](#); [Kalin & Weisser \(2025\)](#)), prosodic constituents are being built bottom-up as well and become accessible only at a given point in the derivation. Each instance of Vocabulary Insertion and Linearization will include a minimal cycle of phonology and prosodification before the next, higher Vocabulary item is inserted.

(1) Architecture of the syntax-PF interface



Going from there, we present two case studies from two radically different empirical phenomena where this property makes the precisely the right predictions. And even though both of these case studies involve different empirical domains of the morphosyntax-prosody mapping, the abstract pattern that they instantiate can be summarized as follows:

(2) Correlation of Syntactic Height and Prosodic Size (CoSHAPS):

If two syntactic heads (or their respective exponents) trigger the same prosodic process on an element X, then the process of the syntactically higher head will apply to a bigger (or equal) prosodic domain within X.

In both of these case studies, we contrast two syntactic heads, which are very local to each other and trigger the same process that refers to prosodic domains (partial vs full

¹This dichotomy is of course, an oversimplification. A number of recent approaches of indirect mapping approaches have begun to incorporate syntactic locality domains (i.e. phases) into the computation of prosodic domains. See e.g. [Ishihara \(2007\)](#); [Kahnemuyipour \(2004, 2009\)](#); [Kratzer & Selkirk \(2007\)](#); [Revithiadou & Spyropoulos \(2009\)](#); [Šurkalović \(2016\)](#).

reduplication or prosodically determined second position effects of clitics). Crucially, even though these heads are syntactically quite local to each other, they still differ with respect to their position above or below a cyclic node. And in both cases, we find crosslinguistically robust generalizations that process triggered by the higher syntactic head can refer to a bigger prosodic constituent than the process triggered by the lower syntactic head.

What we take this to mean is that the bigger prosodic constituent is only constructed (and therefore becomes accessible) upon hitting the cyclic node. This, in turn, means that prosodic structure must be built cyclically and in a bottom-up way. We argue that non-cyclic, representational theories cannot account for these phenomena in a principled way.

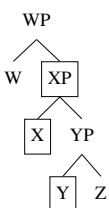
The paper is structured as follows: In Section 2, we will devise a cyclic architecture of the syntax-prosody mapping by laying out the architecture of the system and going through a number of concrete derivations step by step. Sections 3 and 4 will then present the two case studies, which, we argue, make a strong case for the model we develop. Section 5 briefly comes back to the question of Direct vs Indirect Mapping and, in addition, sketches some promising predictions of our model. Section 6 concludes.

2. A cyclic Architecture for prosodification

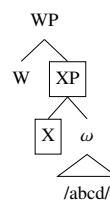
In this section, we will develop a detailed model according to which the mapping from syntax to prosody applies in a strictly cyclic, bottom-up fashion. Each cycle is triggered by an instance of Vocabulary Insertion.

Our basic assumptions are the following: After all structure has been built in syntax, each terminal node must undergo spell out. This consists of two steps. The correct vocabulary item is inserted and the item is linearized with respect to previously inserted structure. These steps take place for every terminal node. A subset of nodes, called cyclic nodes here, introduce an additional step. The previously generated structure is optimized phonologically based on constraints and translated into prosodic structure as part of the optimization. The relevant nodes for the two case studies presented here, are v , V and CP . When such a cyclic node is spelled out, mapping constraints apply to it, following match theory (Selkirk, 2011). Before moving on to the concrete case studies, let us illustrate the concrete mechanism of prosodification through the hypothetical syntactic structure in (3), in which cyclic nodes are marked as boxed. Vocabulary insertion occurs bottom-up, from node to node. Therefore, after all structure was built in syntax, the first node to be spelled out is Z in the structure in (3). The correct vocabulary item is inserted. The next node to be spelled out is Y . Y is assigned the correct vocabulary item and is linearized with respect to the previously spelled out structure of Z . In the hypothetical structure in (3), Y is a cyclic node, which is indicated by the node being boxed. Therefore, the structure that was linearized so far is now optimized and prosodified in phonology. In this case, this results in a prosodic word. The result of these first steps can be seen in (4).

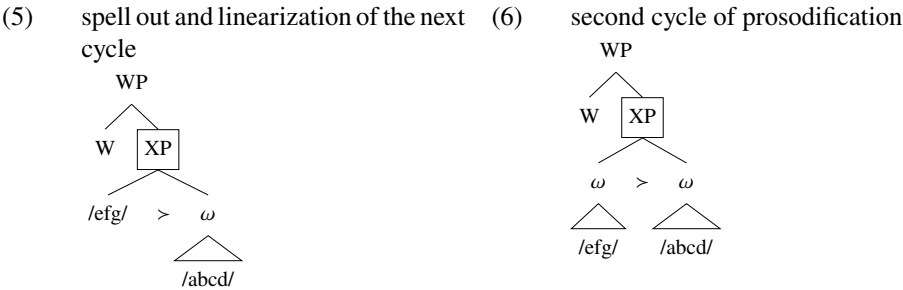
(3) structure in the syntax



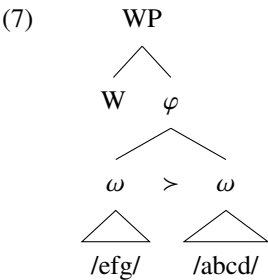
(4) first cycle of prosodification



The next node to be spelled out is X. The correct vocabulary item is inserted and linearized with respect to the already linearized structure. This is seen in (5). Since X is a cyclic node as well, the whole structure is optimized (and therefore also prosodified) again. The result of these steps is illustrated in (6).



Since XP is a cyclic node as well, the whole structure is optimized and prosodified again. This time into a prosodic phrase. See the final structure in (7).



We are now in a position to explain why the size of a domain a phonological process operates over should depend on the height of the head introducing it. When X is spelled out (seen in (5)), there is no prosodic phrase any spellout could interact with. Only prosodic words and lower prosodic structure is available. On the other hand, when a higher head like W is spelled out it has access to the the prosodic phrase that was created in the last cycle of spell out. If the heads X and W in the structure above were spelled out by the same type of prosodic process (for example reduplication or cliticization) this process could therefore only interact with the prosodic structure that is already built at the point of spellout of X or W. This means, X could only interact with a prosodic word or lower (e.g. reduplicate such a prosodic unit) and W could interact with (e.g. reduplicate) a prosodic word or a prosodic phrase. This leads to the generalization that we call CoSHAPS and which is given in (8).

- (8) Correlation of Syntactic Height and Prosodic Size (CoSHAPS):
If two syntactic heads (or their respective exponents) trigger the same prosodic process on an element X, then the process of the syntactically higher head will apply to a bigger (or equal) prosodic domain within X.

In the following we are going to illustrate this generalization with two unrelated phenomena: flavours of verbal number and shifting coordinators. In both cases the same prosodic process

applies to differently big prosodic units depending on the height of the head triggering this process.

3. Application: Two types of verbal number

Event number grammatically marks the number of events. This comes in two flavours called event number and argument number.² An example of each of the flavours can be seen in the examples in (9), coming from Kipsigis, an Eastern-Sudanic language spoken mainly in Kenya.

(9) Kipsigis (Kouneli, 2020, 7)

- a. labat -i là:kwè:t.
run.SG-IPFV child.NOM
'The child is running.' (baseline)
- b. ruaj là:gô:k.
run.PL-IPFV children.NOM
'The children are running.' (Argument number)
- c. ruaj~ruaj là:gô:k.
run.PL~RED children.NOM
'The children are running over and over again.' (event number)

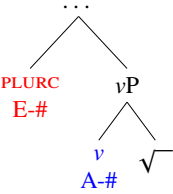
The contrast of (9-a) and (9-b) illustrates argument number. The verb form varies according to the number of the argument. With a singular argument like là:kwè:t 'child' in (9-a) the verb form labat is used to express the verb 'to run'. In contrast to that, a completely unrelated (i.e. suppletive) verb form is used to express 'to run' in (9-b) in context of the plural argument là:gô:k 'children'. In (9-c) now, the plural verb form is reduplicated to express event number, the second flavour of verbal number. This flavour is not concerned with the number of argument/participants the event is distributed over, but instead with the number of times the event takes place. The multiplicity of events, translated by the English phrase 'over and over again' is marked by the total reduplication of the verb form in Kipsigis. Hence, the contrast of (9-b) and (9-c) illustrates event number.

The two flavours of verbal number have been argued to be located in different positions in the syntax. Argument number is argued to be located very low in the verbal domain. See Amato (2018) and Fiebig (in preparation) for approaches where argument number is located on little v and (Thornton, 2017, 2020) and Fiebig (in preparation) on how the argument number head and the verbal root are in a sisterhood relationship. From this very low position, the argument number head agrees with a DP (usually the object) to find the relevant number features. Event number, on the other hand, is located on a head at least outside of vP, called PLURC here.³ See the structure in (10) for a visual representation of the locations of event number (E-#) and argument number (A-#).

²Usually, the term *participant number* is used instead of *argument number*, see for example Corbett (2000) and Thornton (2020). We changed the terminology to better contrast with the term partial reduplication and both terms can be understood as being interchangeable.

³PLURC stands for *Pluractionality*, a different name used for event number, see e.g. Wood (2007).

(10)



Cross-linguistically, event- and argument number are often marked by reduplication (Mattiola, 2019). See (11) for an example from Niuean where both, event- and argument number are marked by reduplication.

(11) Niuean (Haji-Abdolhosseini et al., 2002)

- a. kua ha-hala e ia e tau lā akau.
PERF RED.ARG-CUT ERG he ABS PL branch tree
'He cut the branches.'
(Argument number)
- b. ne noko-noko e ia e gutuhala
PST RED.EVENT-knock ABS she ABS door
'She knocked on the door (many times).'

As can be seen from (11) though, event- and argument number are marked by different types of reduplication. Partial reduplication marks argument number and total reduplication marks event number. In (11-a), the verb *hala* 'to cut' is partially reduplicated (the first mora is repeated) to express argument number. In (11-b), on the other hand, the whole verb *noko* 'to knock' is repeated (i.e., totally reduplicated) to express multiple events. This correlation between the flavours of verbal number and the size of reduplication can not only be observed in Niuean but is attested cross-linguistically. Fiebig (in preparation) investigates 115 languages from 37 families with respect to reduplication as a marker for verbal number. The results of this study can be found in Table 1.

	partial reduplication	full reduplication	unspecified reduplication
event number	28	17	27
argument number	25	0	0

Table 1. reduplication as a marking strategy for verbal number.

The table reveals a typological gap. Event number can be marked by partial or full reduplication cross-linguistically as indicated in the first row of the table.⁴ Argument number, on the other hand, is only marked by partial reduplication throughout the sample. This observation is only limited to reduplication as a spell-out of verbal number though. This makes no statement about other marking strategies for verbal number or other grammatical categories expressed by reduplication. This observation can be summarized as in (12).

⁴Some languages in the sample are not specified for full- or partial reduplication. All of these languages (27) express event number by that unspecified type of reduplication.

(12) Generalization:

Partial reduplication expresses event- and argument number, but full reduplication only ever marks event number. Full reduplication never marks argument number.

This typological gap will find an explanation in the CoSHAPS which is repeated here again in (13).

(13) Correlation of Syntactic Height and Prosodic Size (CoSHAPS):

If two syntactic heads (or their respective exponents) trigger the same prosodic process on an element X, then the process of the syntactically higher head will apply to a bigger (or equal) prosodic domain within X.

Since event and argument number are located in differently high positions in the syntax, the domains reduplication can apply to in each case differs. The higher head (event number) can trigger reduplication of a bigger prosodic domain (prosodic word), while the lower head (argument number) can only trigger reduplication of smaller prosodic domains like moras. This is rooted in the mechanism of reduplication assumed in this analysis. Following [Saba Kirchner \(2010\)](#), we assume that reduplication is one of several competing repair strategies available in phonology to integrate floating elements. Depending on the ranking of constraints like MAX, DEP and INTEGRITY, one of the available strategies (e.g. epenthesis or reduplication) turns out to be the optimal candidate when a floating element is present in the input. When INTEGRITY is ranked lower than DEP, for example, reduplication may be the more optimal way to associate a floating prosodic element with segments than epenthesis. In the following, we are going to show in detail how this approach to reduplication, coupled with our architecture of spell out leads to the typological gap observed in this section.

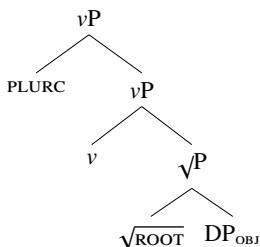
3.1. Full reduplication expresses event number

The first pattern is full reduplication expressing event number. Consider again the Niuean example from above which illustrates this pattern, repeated here as (14).

- (14) ne **noko-noko** e ia e gutuhala
 PST RED.EVENT-knock ABS she ABS door
 ‘She knocked on the door (many times).’

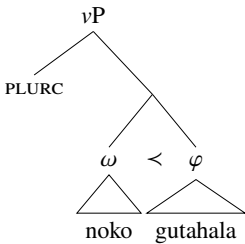
The structure in (15) represents the syntax for this sentence and is therefore the starting point of the derivation.

(15)

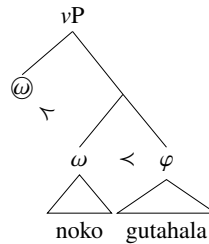


As explained in Section 2, spell-out proceeds bottom for each nodes and includes vocabulary insertion, linearization and optimization. Let us assume that this has happened already for almost all nodes within the structure in (15) including the DP (and all structure within it), the root and little *v*. Since cyclic nodes have been spelled out, the previous spellout steps already included steps of optimization and prosody mapping. Most relevantly, *v* is a cyclic node. Therefore, its spellout is accompanied by a step of optimization and prosodification of the *vp* material into a prosodic word. The result of this step can be seen in the structure in (16). Now follows the next (and most crucial) step of spell out.⁵ The PLURC node receives a vocabulary item (a floating prosodic word) and is linearized with respect to the previously generated structure. See (17) for an illustration of this step.

(16)



(17)

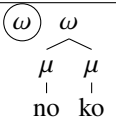
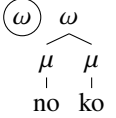
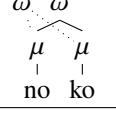
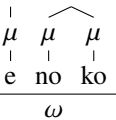
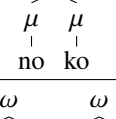
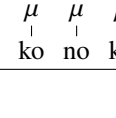


Once the next cyclic node is spelled out (*V*, not shown here), the next step of optimization takes place. This is where the floating element is taken care of and as a result, reduplication arises. Consider the tableau in (18) which illustrates the optimization step.⁶ The tableau illustrates five relevant constraints. *FLOAT assigns a violation for floating material in the output structure; NOCROSS(ING ASSOCIATION LINES) forces the output to avoid crossing association lines; DEP assigns violations for all material that is in the output and not in the input and MAX does the reverse. INTEG(RITY) assigns violations for “each segment in the input with more than one correspondent in the output.” (McCarthy & Prince, 1995). Candidate a. is the faithful candidate and is excluded by the highest ranked constraint which hinders floating elements. Candidate b. shows a possible repair strategy for the floating element: integration of the already existing material into the floating prosodic word. But this violates NOCROSS(ING ASSOCIATION LINES) and candidate b. is therefore also not optimal. Candidates c. and d. show further possible repair strategies: adding material to fill the floating prosodic word or deleting the floating element all together. These candidates are not optimal due to the DEP and MAX constraints. The winning candidate is e. which includes a reduplication as repair strategy for the floating element. This reduplication violates INTEG(RITY) by having two segments in the output that correspond to one segment in the input. But since this constraint is ranked lower than all other relevant constraints, this violation is not fatal. Therefore, reduplication of the prosodic word that was spelled out earlier is the output of optimizing the structure in (17).

⁵There might have been a step of (head)movement that took place prior to this spell out operation as is evident from the surface order in (14). We are not discussing this here since the phonological context relevant for the spell out of *vP* remains the same, wherever it might take place.

⁶The input structure is somewhat simplified and only shows the relevant part of the structure, the floating element and the prosodic word right next to it. The tableau also doesn't show the step of prosodification that takes place in this cycle.

(18) Full reduplication for event number

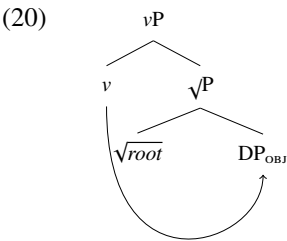
I:		*FLOAT	NoCROSS	DEP	MAX	INTEG
a.		*!				
b.			*!			
c.				*!		
d.					*!	
e.						****

3.2. Partial reduplication expresses argument number

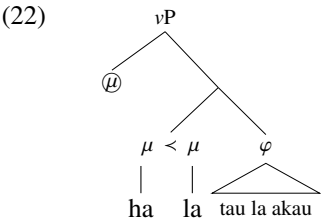
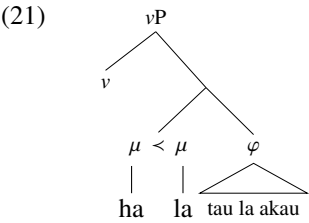
For partial reduplication expressing argument number, consider again the Niuean example repeated here as (19).

- (19) kua ha-hala e ia e tau lā akau.
PERF RED.ARG-CUT ERG he ABS PL branch tree
'He cut the branches.'

This sentence is structurally represented in (20). (20) also indicates the agreement which takes place between little *v* and the object DP. This agreement values the probe (little *v*) as plural and leads to the spell out of argument number on the verb.



After the structure has been built in the syntax and agreement processes have happened, spell out takes place. Let us assume, again, that the DP has already been spelled out and mapped onto a prosodic phrase. The next terminal node to be spelled out is the root. The correct vocabulary item, including its mora structure is inserted and linearized with respect to the already spelled out prosodic phrase. This can be seen in (21). Crucially, the root has not been prosodified at this point, since the relevant cyclic node (little *v*) has not been spelled out yet. Only in the next step, little *v* is now spelled out and linearized with respect to the previously spelled out structure. Since *v* was valued for plural by agreement, the right vocabulary item is a floating mora. It is inserted and linearized, as shown in (22).



Since *v* is a cyclic node, a step of optimization and prosodification now follows. The spelled out verbal structure needs to be mapped onto a prosodic word. This optimization step can be seen in (23). In this tableau, it is not a constraint against crossing association lines, but instead a constraint **AMBIMOR* against ambimoraic elements that stops the floating element from simply associating with the already existing material. Again, reduplication arises as the most optimal repair for the floating element. Crucially, reduplication alone is not enough. The winning candidate is not e., but f. This is because of the constraint *MATCH*(*X*⁰,*ω*). This constraint ensures that the correct prosodification of the node *v* to a prosodic word takes place in this optimization step.

(23) Partial Reduplication for argument number

I:	  						
	  	<i>*FLOAT</i>	<i>*AMBIMOR</i>	<i>DEP</i>	<i>MAX</i>	<i>INTEG</i>	<i>MATCH</i> (<i>X</i> ⁰ , <i>ω</i>)
a.	  	<i>*!</i>					<i>*</i>
b.	  		<i>*!</i>				<i>*</i>
c.	  			<i>*!</i>			<i>*</i>
d.	  				<i>*!</i>		<i>*</i>
e.	  					<i>**</i>	<i>*!</i>
f.	   					<i>**</i>	

As was shown in the Table in 1, partial reduplication can also express event number. This process would be almost identical to the one described above for argument number. The only difference would be that the higher PLURC head is spelled out by a floating mora in this case. In the same way as in (23), this floating mora triggers reduplication of a mora (under violation of INTEGRITY) to satisfy *FLOAT.

3.3. Why Full reduplication never expresses argument number

After having seen the previous derivation, the crucial question remains: why is it that argument number can never be spelled out by full reduplication? As we saw above, total reduplication requires a floating prosodic word that is then repeated as part of an optimization step to not violate a *FLOAT constraint. Let us now look at a possible derivation with a floating prosodic word as a spell out of argument number. Assume that all steps of structure building, agreement and spell out have taken place, just as in the previous case in Section 3.2. A floating prosodic word is now inserted as the correct vocabulary item for little *v* and this linearized structure serves as the input to an optimization step illustrated in (24). You can see that the input is identical to that in the previous tableau in (23) besides the floating element that needs to be integrated. This time, reduplication is not the winning repair strategy even though the reduplicated candidate c. violates only the low ranked integrity constraint. Instead, candidate b. is the most optimal. This is the case, because integrating the already existing segments into the floating prosodic word violates no constraints in this scenario. No crossing association lines result from such an integration, as it would have been the case for candidate b. in Tableau (18).

(24) No Reduplication for low @

I: ω μ μ ha la	*FLOAT	*AMBIMOR	NoCROSS	INTEG	MATCH(X^0, ω)
a. ω μ μ ha la	*!				*
b. ω μ μ ha la					
c. ω ω μ μ μ μ ha la ha la				*!***	

The crucial difference between both cases is timing. When argument number is spelled out, no prosodic word has been prosodified yet and therefore the association of the floating prosodic word and the already existing segments is actually cheaper than reduplication. In the earlier cases, reduplication was the cheapest option, only since integration (and other repair strategies like epenthesis (24)) were more costly due to higher ranked constraints. This observation is able to explain the typological gap introduced earlier: Argument number is never spelled out by total reduplication. Since input and output of the optimization step in (24) look identical on the surface, we can never observe such a process. This means, that if

a language would have a configuration that could potentially lead to total reduplication with argument number (a floating prosodic word spelling out a plural valued little v), it would never actually lead to total reduplication. This case study is an instance of the CoSHAAPS. The height of a syntactic element (argument number vs. event number) influences the size of the prosodic domain (ω vs. μ) a phonological process operates over (reduplication) that is triggered by the syntactic elements.

4. Application: Two types of coordinators

In this section, we turn to another empirical phenomenon which illustrates the workings of the CoSHAAPS, namely second position effects of clausal coordinators.

4.1. Monosyndetic and Polysyndetic coordinators

Just like we compared two flavors of verbal number in the previous section, this section compares two flavors of clausal coordination. In the relevant literature, these flavors are known as monosyndetic and polysyndetic coordination patterns (see e.g. [Bhatt 2014](#); [Haspelmath 2007](#); [Mitrović & Sauerland 2014](#)) and they are the two main morphosyntactic encodings of coordination structures. Morphologically, the two strategies can be distinguished by the number of linking elements (coordinators). Monosyndetic coordination makes use of one coordinator, which is located in between the two conjuncts. Polysyndetic coordination is characterized by having one coordinator per coordinand. Polysyndetic coordinators are usually suffixed (or enclitic) to the respective coordinand even though prefixed (or proclitic) polysyndetic coordinators are occasionally attested too. The following representations illustrate the two canonical positions.

(25) The canonical position of a monosyndetic coordinator:

A < & < B

(26) The canonical positions of polysyndetic coordination:

a. A < & < B < &

b. & < A < & < B

The two patterns are illustrated with data involving nominal coordination from Udmurt in (27):

- (27) a. Petyr-en Maša-jen
Peter-INS Masha-INS
b. Petyr no Maša
Peter and Masha
'Peter and Masha'

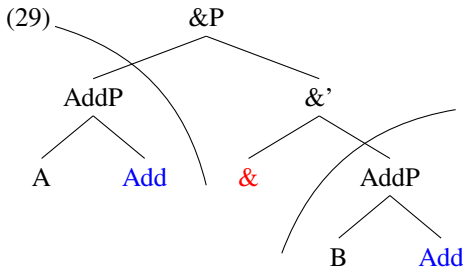
[Weisser \(2017\)](#)

It has been observed that in languages, in which both strategies are present, they can usually be combined.

- (28) Petyr-en no Maša-jen
Peter-INS and Masha-INS
'Peter and Masha'

In line with the observation that these elements are not in complementary distribution, Bhatt (2014) and Mitrović & Sauerland (2014) have proposed that these elements do not occupy the same position in the structure and that polysyndetic coordinators attach to the individual conjuncts (Add in (29)) and that it is the monosyndetic coordinator that links the conjuncts (& in (29)). Languages can then pick and choose which one of these heads is overt and which is covert.⁷

In addition, it has been noticed that while monosyndetic coordinators only have one syntactic function, namely to coordinate constituents, polysyndetic coordinators often have a number of focus-related functions such as additive focus (often translated as ‘also’), scalar focus (translated as ‘even’) or that they mark contrastive topics (for a more elaborate discussion, see Forker 2016). Following Bhatt (2014), it is thus plausible to assume that these additive focus markers are actually part of the extended CP-domain. Given the assumption that the individual conjuncts in coordination are phases (or cyclic domains),⁸ the following picture emerges, where the monosyndetic coordinator & is immediately outside of the cyclic domain whereas the polysyndetic one Add is just inside of it:



On an abstract level, we see that the syntactic configuration is reminiscent of the one in the previous section for verbal number. One type of coordinator is right below a cyclic node and the other type of coordinator is right above it.

4.2. Second position effects

The next ingredient we need is the morphophonological process that references cyclic domains. In this case, the process in question is one of prosodically determined second position placement. It has long been known that some elements appear in a second position inside of the domain that they take scope over (see amongst many others Anderson 2005; Klavans 1995; Wackernagel 1892; Zwicky 1970). In some cases, this second position is defined in syntactic terms, in which case the clitic follows the first syntactic phrase. However, in other cases, this second position is defined in prosodic terms in which case the clitic follows the first prosodic word *or* the first prosodic phrase. It is the latter cases, which we are interested in.

⁷It should be mentioned that polysyndetic coordinators should not be distinguished purely based on their number in a concrete example because in a number of cases, languages making use of the polysyndetic strategy can drop coordinators in one of the conjuncts. In the Latin example below, *=que* counts as a polysyndetic coordinator even though it occurs only once. As argued for in detail in Weisser (2024), the decisive criterion is whether they can appear on every conjunct, not whether they do in a concrete example. As for example Mitrović (2014) shows, Latin allows for doubling of *que* in every conjunct, cf. *noct-es=que die-s=que* ‘nights and days’ Cicero, *De Finibus*, 1,16,51.

⁸See e.g. Bošković (2020); Weisser (2015)

Coordinators such as the Latin *-que* have always featured as particularly convincing candidates for prosodically determined second position placement as their scope can quite readily be read off the surface structure. In the example below, two clauses are conjoined with *=que* but its surface position is after the first prosodic word of the second conjunct.

- (30) [Is istum reliqui-t] [{de provincia}_ω=*que* decess-it]
 he it leave.PERF-3SG from province=AND depart.PERF-3SG
 ‘He left it and departed from the province.’ Cicero, Against Verres 2.2.48

Weisser (2024) presents a qualitative typology of such cases of non-canonical coordinator placement and finds that these coordinators exhibit the full range of second position placement types argued for in the literature. Of particular interest to the discussion at hand are cases such as the one from Yorùbá where, as has been argued in detail by Aremu & Weisser (2024), the coordinator *sì* attaches to the first prosodic phrase of the second conjunct.

- (31) ... {Olá ò bá}_φ *sì* lọ.
 ... Ola should have COORD go
 ‘...and Ola should have gone.’ Yorùbá, Aremu & Weisser (2024)

Latin *=que*, which is a polysyndetic coordination marker, attaches to the first prosodic word and Yorùbá *sì*, which is a monosyndetic coordination marker attaches to the first prosodic phrase. As shown in the table in (32), this dichotomy carries over to all languages discussed in the sample in Weisser (2024):

	[ω -COORD	[ϕ -COORD
(32) <i>ADD</i>	Latin, Cherokee, Kalaallisut, Ancient Greek Hittite, Gaulish, Gothic, Old Irish, Coptic	–
<i>&</i>	–	Yorùbá, German, Polish

Against the background of the trees above, it is not hard to see that this correlation can be made sense of in terms of the CoSHAPS. The higher syntactic head *&*, which is just outside of the cyclic CP-node, has access to the bigger prosodic category ϕ . The lower syntactic head underneath the cyclic CP-node does not.

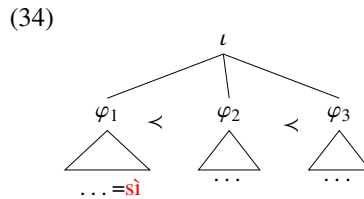
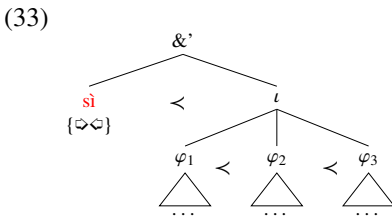
4.3. Deriving the correlation

In this subsection, we will briefly sketch a model that derives the correlation between the two types of coordinators and the prosodic category they attach to. This sketch is based on the Distributed Integration model for clitic placement developed in Weisser (2024). The model has the following properties: (a) It is transformational, which means that the clitic is base-generated in its semantically transparent position and arrives at its surface position through a transformation. (b) It is subcategorization-based, which means that elements that undergo the clitic-placement transformation are lexically specified to appear in certain position. It is (c) cyclic, which means that the transformation applies as soon as possible once the subcategorization-feature has entered the structure. (d) It is distributed, by which we mean that the transformation can apply during Spell-Out (i.e. during the mapping from syntax to morphology) or during Vocabulary Insertion (i.e. during the

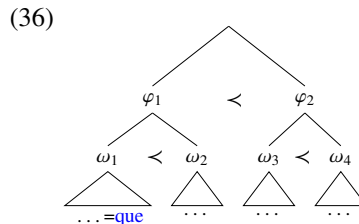
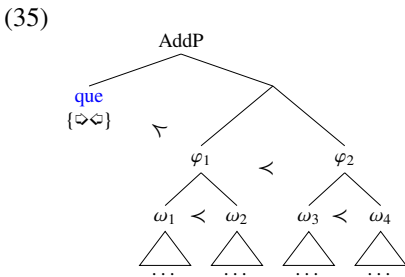
mapping from morphology to phonology). The latter assumption derives the dichotomy of morphosyntactically determined clitic placement and prosodically determined clitic placement and thus, only the step of Vocabulary Insertion is relevant.

The transformation in question is called **INTEGRATION** in Weisser (2024) and, when it applies during Vocabulary Insertion, it makes reference to the linearization statements discussed in Section 2. What the transformation does is that instead of attaching them directly to the element they precede, it attaches them to the first element *within* the element they precede.⁹

The tree in (33) shows how this applies to a monosyndetic coordinator. This coordinator takes a CP-complement. Due to the cyclic prosodification algorithm sketched in Section 2, the internal structure of the CP is already prosodified and mapped to a collection of φ -phrases as seen in (33). The CP itself, being a cyclic node, is mapped to an intonation phrase ι . As soon as the Vocabulary Insertion procedure arrives at the coordinator, its Vocabulary Item (here, the Yorùbá exponent *sì*) is inserted and linearized with respect to the previously generated structure (i.e. wrt. ι). Since the exponent comes with a subcategorization feature (here depicted as $\{\triangleright\varphi\}$ ¹⁰), it will undergo the above-mentioned **INTEGRATION** operation. This operation will right-adjoin it to what is linearly the first element of the element it precedes ($<$). The exponent *sì* precedes ι and within ι , φ_1 is the first element. Then, cyclic phonology applies leading to proper phonological integration of *sì* into φ_1 (34). A clausal monosyndetic coordinator will always attach to the first φ -phrase because it sits right above the cyclic node (CP) that closes off the intonation phrase.



A polysyndetic coordinator however sits below the cyclic node, which means that while the elements within its complement have mapped to a collection of φ -phrases as above, the complement itself has not yet been mapped to an intonation phrase. So, as soon as Vocabulary Insertion hits the polysyndetic coordinator, its exponent (here the Latin *que*) is inserted and linearized with respect to the previously generated structure. This, in this case, however, means that it will be linearized with the first prosodic phrase φ_1 .



⁹The formal definition in Weisser (2024) abstracts over relations of grammatical constituents (such as linearization and c-command) such that it applies to hierarchical and linear structure alike.

¹⁰See Weisser (2024) for more explanation on the radically underspecified nature of this subcategorization feature.

Thus INTEGRATION will dislocate the coordinator to a position within the first prosodic word (ω_1) because the coordinator precedes ($<$) φ_1 and within φ_1 , ω_1 is the first element. The polysyndetic coordinators appear after the first word because the respective syntactic head sits right underneath the cyclic node CP. Only upon hitting the CP-node, the intonation phrase is created which leads to a distinct second position pattern.

5. Extensions and Predictions

The current account predicts that the correlation of syntactic height and prosodic size should not be restricted to shifting coordinators and verbal number reduplication. Instead, it should apply to all morphosyntactically conditioned processes that make reference to prosodic constituents, i.e. the prosodic word, the prosodic phrase and the intonational phrase. In this subsection, we will discuss two such areas: truncation and subcategorization frames for affixes.

Truncation has been one of the foundational topics of Prosodic Morphology. [Alber & Arndt Lappe \(2012\)](#) define templatic truncation as a process where a morphological category is typically marked by the lack of material in the derived form that is present in the base form and the size — in terms of syllables, feet, or higher units — of the derived form is predictable. Take the Russian hypocoristics of proper names in (37) and the clippings of common nouns in (38). The base forms have a diverse syllabic and prosodic structure whereas the truncated forms all consist of a trochaic foot of the form CVCV.

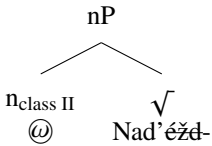
(37) Russian hypocoristics (Steriopolo, (38) 2018, 58)	Russian clippings (Steriopolo, 2018, 66)
a. Bor'ís Boris.M.NOM.SG.I 'Boris (full form)'	a. p'etúx rooster.M.NOM.SG.I 'rooster (full form)'
b. Bór'-a Borja-M.NOM.SG.II 'Borja (hypocoristics)'	b. p'et'-a rooster-M.NOM.SG.II 'rooster (colloquial)'
c. Nad'ěžd-a Nad'ežda-F.NOM.SG.II 'Nad'ežda (full form)'	c. pár'en guy.M.NOM.SG.I 'guy (full form)'
d. Náď'-a Nadja-F.NOM.SG.II 'Nadja (hypocoristics)'	d. pár'-a guy-M.NOM.SG.II 'guy' (colloquial)

The predictions of our approach become evident if the classic Prosodic Morphology ([McCarthy, 1999](#)) approach is translated into autosegmental representations in parallel to [Saba Kirchner \(2010\)](#) Minimal Reduplication. Truncation would then be triggered by a floating word that potentially already dominates other material.¹¹ Under the assumption that these templates cannot simply overwrite existing prosodic structures, these truncating templates can only have an effect if they are added before other prosodic structure is built. For the Russian example, we would need to assume that there is a prosodic word introduced

¹¹Further evidence for such a move comes from the idea that truncations often correspond to minimal stems, words, or roots in several languages ([Downing, 2006](#); [Krämer, 2018](#)).

by a categorizing head *n*, which was previously attached to a root, which is exponed by the proper name *Nad'ěžd-*, without any inflection or inflection class features. Integrating its segmental material into the prosodic word will yield the optimal prosodic word, which in Russian is monosyllabic (Gouskova, 2019, 175).

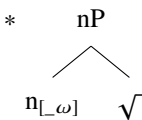
(39) Example Structure of Russian truncation



Integrating the segmental material and deleting any left-over material is easily possible if the truncation-triggering exponent is inserted into a categorizing head *n*. This is exactly what is argued for by Steriopo (2018), who shows that hypocoristic of proper names and clippings of common nouns in Russian, cf. (38), all show the same inflection class — a feature that is assumed to be located on little *n*.

The second area concerns affixes that are not integrated into a prosodic word but instead adjoin to it, such certain German suffix classes (Raffelsiefen, 2000) or clitics in Italian varieties (Peperkamp, 1996). If these affixes are analyzed using subcategorization frames (such as the conditions on insertion in (Kalin & Rolle, 2023), we would expect these to be in a morphosyntactic position that is higher than the first categorizing head. If they were to be added before the formation of prosodic words, their prosodic subcategorization frame could not be satisfied. From a cursory glance on the literature, this seems to be true for the reported data.

(40) Impossible Subcategorizing affix



A related prediction of cyclic prosodification is discussed in Tebay (to appear). Certain exponents in Indonesian attach outside of the stress domain, i.e. outside of a prosodic phrase. This extraprosodic positioning leads to an exceptional stress pattern and can be made opaque in later cycles in two ways. First, certain other, non-phrasal affixes, can attach in a later cycle. Since prosodification is cyclic, this leads to a loss of extraprosodicity of the exponent under the assumption that only a single element can be extraprosodic (Hayes, 1982). Second, in the postcyclic phonology, certain phonological processes treat the exponent as part of the prosodic word, which is of course unexpected in most parallel frameworks of extraprosodicity. Under a cyclic prosodification approach, however, the cyclic structure can be altered after stress has been assigned, which leads to different prosodic domains for postcyclic processes. The framework thus predicts that subcategorization frames only need to be satisfied at the cycle where the affixes are introduced but not if the prosodic structure is later altered for independent reasons.

6. Conclusion and Outlook

In this paper, we have proposed a cyclic model of prosodification, where prosodification and phonological optimization applies at each cyclic node to all vocabulary item inserted and linearized at this point. We have provided evidence from verbal number reduplication and shifting coordinators for the central property of this approach – the correlation of syntactic height and prosodic size. In both areas, operations triggered by an exponent inserted into higher syntactic heads can refer to larger prosodic constituents than processes triggered by a lower exponent.

The combination of cyclicity and prosody in the present approach raises an interesting question. Could any of the two ingredients single-handedly derive the correlation of syntactic height and prosodic size – potentially even in a more elegant way? We think the answer is no, and will explain why in the rest of this section.

Prosodic constituents are often combined with a parallel framework of grammar, such as standard parallel optimality theory (cf. e.g. [Nespor & Vogel, 1986](#)). In such a system, the correlation between syntactic height and prosodic size cannot follow from cyclicity. Instead, it would have to be stipulated as a restriction on the correspondence between morphosyntactic structure and prosodic structure. Crucially, these constraints would need to be introduced in addition to constraints on prosodic mapping. [Ito & Mester \(2000, 33\)](#) introduce a constraint labeled Hierarchical Locality, which restricts access of a condition operating on prosodic structure to two hierarchically adjacent levels. Note that such a constraint is not sufficient since it only relates prosodic levels. A stipulated locality constraint that corresponds to the correlation of syntactic height and prosodic size would need to link syntactic substance and prosodic size instead.

In a framework based purely on cyclic evaluation, where all domains relevant for prosodic processes are fully isomorphic with morphosyntactic domains, specific problems would arise whenever the domains are not in fact isomorphic. This is the case for clitic placement in Yoruba ([Weisser, 2024](#)). The position of the coordinator *sì* can be described as following the first prosodic phrase but it cannot be described in simple syntactic terms.

- (41) Olú ti rà as'ò obinrin [tí Adé **sì** ri ní anà] ti ra
 Olu PERF bought 'clothes woman that Ade CONJ saw at yesterday PERF buy
 bàtà
 shoes
 'Olu has bought clothes and the woman who Adé saw yesterday has bought shoes'

As shown in the example in (41), it follows the subject of a relative clause that modifies the subject of the coordinated clause. Syntactically, this means that the coordinator would need to follow a constituent consisting of the modified noun and the modifying relative clauses subject or occur inside the modifying relative clause below the subject position. Neither of these offers a simple syntactic explanation of the clitic placement here. Prosodically, on the other hand, it can be described as following the first prosodic phrase.

A further problem of a cyclic non-prosodic approach concerns the unity of the effects of the correlation of syntactic height and prosodic size. If there were in fact no prosodic constituents, then why should clitic placement and reduplication show the same effects? If we were to assume a purely syntactic theory of clitic placement (e.g. [Bošković, 2001](#)) and a non-prosodic theory of reduplication, e.g. Morphological Doubling Theory ([Inkelas & Zoll,](#)

2005), then why should these two unrelated theoretical devices result in parallel restrictions on possible patterns? In Morphological Doubling Theory, which is — as far as we are aware — the only theory of reduplication that does not need to resort to prosodic constituents for partial and full reduplication, reduplication is a compounding construction, which requires identity between compounds and potentially restricts the phonological size of one of its members. In such a theory, a correlation between the reading of verbal number and the kind of reduplication would need to be stipulated. Why should phonological restrictions in a specific reduplication construction be constrained by a generalization that also holds true for syntactic clitic placement? The present cyclic prosodic account of the data allows for a unified explanation, which can be reduced to the effects of cyclic prosodification.

References

- Alber, Birgit & Sabine Arndt Lappe. 2012. Templatic and subtractive truncation. In Jochen Trommer (ed.), *The morphology and phonology of exponence*, 289–325. Oxford University Press.
- Amato, Irene. 2018. A morphosyntactic account to verbal number in Mupun. In *Proceedings of the 26th conference of the student organization of linguistics in Europe*, 20–44.
- Anderson, Stephen R. 2005. *Aspects of the theory of clitics* Oxford studies in theoretical linguistics. Oxford: Oxford University Press.
- Aremu, Daniel & Philipp Weisser. 2024. Prosodically determined coordinator placement in Yorùbá. *Glossa: A journal of general linguistics* 9(1).
- Bhatt, Rajesh. 2014. Coordination and finiteness in Malayalam: A commentary on Jayaseelan (2013). *Natural Language & Linguistic Theory* 32(1). 213–229. Special Issue: Finiteness in South Asian Language.
- Bobaljik, Jonathan. 2000. The ins and outs of contextual allomorphy. In K. Grohmann & C. Struijke (eds.), *University of maryland working papers in linguistics*, 35–71.
- Bobaljik, Jonathan. 2012. *Universals in comparative morphology: Suppletion, superlatives, and the structure of words*. MIT Press.
- Bošković, Željko. 2001. *On the nature of the syntax-phonology interface: Cliticization and related phenomena*. Amsterdam: Elsevier.
- Bošković, Željko. 2020. On the coordinate structure constraint, across-the-board-movement, phases and labeling. In Jeroen van Craenenbroeck, Cora Pots & Tanja Temmerman (eds.), *Recent developments in phase theory*, 133–182. Berlin: Mouton De Gruyter.
- Cheng, Lisa Lai-Shen & Laura J. Downing. 2016. Phasal syntax = cyclic phonology? *Syntax* 19(2). 156–191.
- Corbett, Greville G. 2000. *Number*. Cambridge: Cambridge University Press.
- D'Alessandro, Roberta & Tobias Scheer. 2015. Modular PIC. *Linguistic Inquiry* 46. 593–624.
- Dobashi, Yoshihito. 2003. *Phonological phrasing and syntactic derivation*: Cornell University dissertation.
- Downing, Laura J. 2006. *Canonical forms in prosodic morphology*, vol. 12. OUP Oxford.
- Elfner, Emily. 2012. *Syntax-prosody interactions in Irish*: Unversity of Massachuesetts Amherst dissertation.

- Elfner, Emily. 2015. Recursion in prosodic phrasing: Evidence from connemara irish. *Natural Language & Linguistic Theory* 33. 1169–1208.
- Embick, David. 2010. *Localism versus globalism in morphology and phonology*. MIT Press.
- Fenger, Paula. 2020. *Words within words: The internal syntax of verbs*: University of Connecticut dissertation.
- Fenger, Paula & Philipp Weisser. accepted. Matching domains: The syntax, morphology and phonology of the verb in sinhala. *Natural Language & Linguistic Theory* ().
- Fiebig, Nadja. in preparation. *The morpho-syntax of verbal number - Verbal number in Kalenjin and beyond.*: Leipzig University dissertation.
- Forker, Diana. 2016. Toward a typology for additive markers. *Lingua* 180. 69–100.
- Gouskova, Maria. 2019. Phonological words in the syntax and in the lexicon: A study of Russian prepositions. *Journal of Slavic Linguistics* 27(2). 161–212.
- Haji-Abdolhosseini, Mohammad, Diane Massam & Kenji Oda. 2002. Number and Events: Verbal Reduplication in Niuean. *Oceanic Linguistics* 41(2). 475–492.
- Haspelmath, Martin. 2007. Coordination. In Timothy Shopen (ed.), *Language Typology and Syntactic Description*, 1–51. Cambridge University Press.
- Hayes, Bruce. 1982. Extrametricality and English stress. *Linguistic Inquiry* 13(2). 227–276.
- Inkelas, Sharon & Cheryl Zoll. 2005. *Reduplication: Doubling in morphology*. Cambridge University Press.
- Ishihara, Shinichiro. 2007. Major phrase, focus intonation, multiple spell-out (map, fi, mso). *The Linguistic Review* 24. 137–167.
- Ito, Junko & Armin Mester. 2000. Weak layering and word binarity. In Takeru Honma, Masao Okazaki, Toshiyuki Tabata & Shin ichi Tanaka (eds.), *A new century of phonology and phonological theory a festschrift for professor Shosuke Haraguchi on the occasion of his sixtieth birthday*, 43–56. Tokyo: Kaitakusha.
- Kahnemuyipour, Arsalan. 2004. *The syntax of sentential stress*: University of Toronto dissertation.
- Kahnemuyipour, Arsalan. 2009. *The syntax of sentential stress*. Oxford University Press.
- Kaisse, Ellen. 1985. *Connected speech: The interaction of syntax and phonology*. Academic Press.
- Kalin, Laura. 2022. Infixes really are (underlyingly) prefixes/suffixes: Evidence from allomorphy on the fine timing of infixation. *Language* 98(4). 641–682.
- Kalin, Laura & Nicholas Rolle. 2023. Deconstructing subcategorization: Conditions on insertion versus conditions on position. *Linguistic Inquiry* 55(1). 197–218.
- Kalin, Laura & Philipp Weisser. 2025. Minimalism and morphology. In Kleanthes Grohmann & Evelina Leivada (eds.), *Cambridge handbook of minimalism*, Cambridge University Press.
- Klavans, Judith. 1995. *On clitics and cliticization: The interaction of morphology, phonology, and syntax*. New York Garland.
- Kouneli, Maria. 2020. Layers of morphosyntactic number in Eastern Sudanic. DFG grant proposal.
- Krämer, Martin. 2018. The perfect prosodic word in Italian: Or fruit salad matters. In Ryan Bennett, Andrew Angeles, Adrian Brasoveanu, Dhyana Buckley, Nick Kalivoda, Shigeto Kawahara, Grant McGuire & Jaye Padgett (eds.), *Hana-bana: a festschrift for junko ito and armin mester*, <https://escholarship.org/uc/item/09m654fp>.

- Kratzer, Angelika & Elisabeth Selkirk. 2007. Phase theory and prosodic spellout: The case of verbs. *The Linguistic Review* 24. 93–135.
- Mattiola, Simone. 2019. *Typology of pluractional constructions in the languages of the world* Typological studies in language. Amsterdam: John Benjamins Publishing Company.
- McCarthy, John & Alan Prince. 1995. Faithfulness and reduplicative identity. In Jill Beckman, Laura Walsh Dickey & Suzanne Urbanczyk (eds.), *Papers in optimality theory*, vol. 18 University of Massachusetts Occasional Papers in Linguistics, 249–384. Amherst, MA: Graduate Linguistic Student Association.
- McCarthy, John J. 1999. Prosodic morphology 1986. *Phonological theory: the essential readings* 238–288.
- Mitrović, Moreno. 2014. *Morphosyntactic atoms of propositional logic*: University of Cambridge dissertation.
- Mitrović, Moreno & Uli Sauerland. 2014. Decomposing coordination. In Jyoti Iyer & Leland Kusmer (eds.), *Proceedings of the NELS 44*, 39–52. Amherst: GLSA.
- Nespor, Marina & Irene Vogel. 1986. *Prosodic phonology*. Dordrecht: Foris.
- Newell, Heather. 2008. *Aspects of the morphology and phonology of phases.*: McGill University dissertation.
- Newell, Heather & Glyne Piggott. 2014. Interactions at the syntax-phonology interface: Evidence from Ojibwe. *Lingua* 150. 332–362.
- Pak, Marjorie. 2008. *The postsyntactic derivation and its phonological reflexes*: University of Pennsylvania dissertation.
- Peperkamp, Sharon. 1996. On the prosodic representation of clitics. In Ursula Kleinhenz (ed.), *Interfaces in phonology*, 102–127. Akademie Verlag.
- Raffelsiefen, Renate. 2000. Evidence for word-internal phonological words in German. In Rolf Thieroff, Matthias Tamrat, Nanna Fuhrhop & Oliver Teuber (eds.), *Deutsche grammatik in theorie und praxis*, 43–56. Niemeyer.
- Revithiadou, Anthi & Vassilios Spyropoulos. 2009. A dynamic approach to the syntax-phonology interface: A case study from greek. In Kleanthes k. Grohmann (ed.) (ed.), *Interphases: Phase-theoretic investigations of linguistic interfaces*, Oxford University Press.
- Saba Kirchner, Jesse. 2010. *Minimal reduplication*: University of California, Santa Cruz Phd dissertation.
- Selkirk, Elisabeth. 1982. *The syntax of words*. MIT Press.
- Selkirk, Elisabeth. 2000. The interaction of constraints on prosodic phrasing. In Gösta Bruce & Merle Horne (eds.), *Prosody: Theory and experiment - studies presented to gösta bruce*, 231–261. Dordrecht: Kluwer.
- Selkirk, Elisabeth. 2009. On clause and intonational phrase in japanese: the syntactic grounding of prosodic constituent structure. *Gengo Kenkyu* 136.
- Selkirk, Elisabeth. 2011. The syntax-phonology interface. In John Goldsmith, Jason Riggle & Alan Yu (eds.), *Handbook of phonological theory*, Oxford: Blackwell 2nd edn.
- Steriopolo, Olga. 2018. Morphosyntactic properties of short first names (hypocoristics) in russian. *Russian Linguistics* 42(1). 57–68.
- Šurkalović, Dragana. 2016. *The no-reference hypothesis: A modular approach to the syntax-phonology interface*: UiT Norges arktiske universitet dissertation.
- Tebay, Sören E. to appear. A prosodic-cyclic theory of circumfix linearization: Evidence from abstract nominalizations in Indonesian. *Linguistic Variation* .

- Thornton, Abigail. 2017. Reduplicating verbal plurals cross-linguistically. In *Berkeley linguistic society proceedings*, .
- Thornton, Abigail. 2020. *Morphophonological and morphosyntactic domains*: University of Connecticut Doctoral dissertation.
- Truckenbrodt, Hubert. 1999. On the relation between syntactic phrases and phonological phrases. *Linguistic Inquiry* 30. 219–56.
- Wackernagel, Jacob. 1892. über ein Gesetz der indogermanischen Wortstellung. *Indogermanische Forschungen* 1. 333–436.
- Wagner, Michael. 2005. *Prosody and recursion*: MIT dissertation.
- Weisser, Philipp. 2015. *Derived Coordination - A Minimalist Perspective on Clause Chains, Converbs and Asymmetric Coordination*. Berlin & Boston: de Gruyter.
- Weisser, Philipp. 2017. Three types of Coordination in Udmurt. In R. A. Kudryavtseva (ed.), *Problems in Mari and Comparative Philologies. Yoshkar-Ola*, .
- Weisser, Philipp. 2024. A typology of shifted coordinators and what they tell us about clitics and their formal modelling. Habilitation Thesis: Universität Leipzig.
- Wood, Esther Jane. 2007. *The semantic typology of pluractionality*. Berkeley: University of California Doctoral dissertation.
- Zwicky, Arnold. 1970. Auxiliary reduction in english. *LingI* 1(3). 323–336.